

MySQL → Sequential Query Language

Data Modeling → It is a critical steps in defining data structure, creating data model to describe associations and constraints for reuse. Design, Organizational structure. it may represent diagrams, symbols or text.

Terminology -

Entity → Entities are the objects or items in the business environment for which we store the data, ^{ex:} ecommerce data - (table Name) → Customers, orders, products.

Table

Attribute → It provide a way to structure and organize the data within an entity. Name, DOB, Address, Email.

Relationship → It is use to resolve the relationship

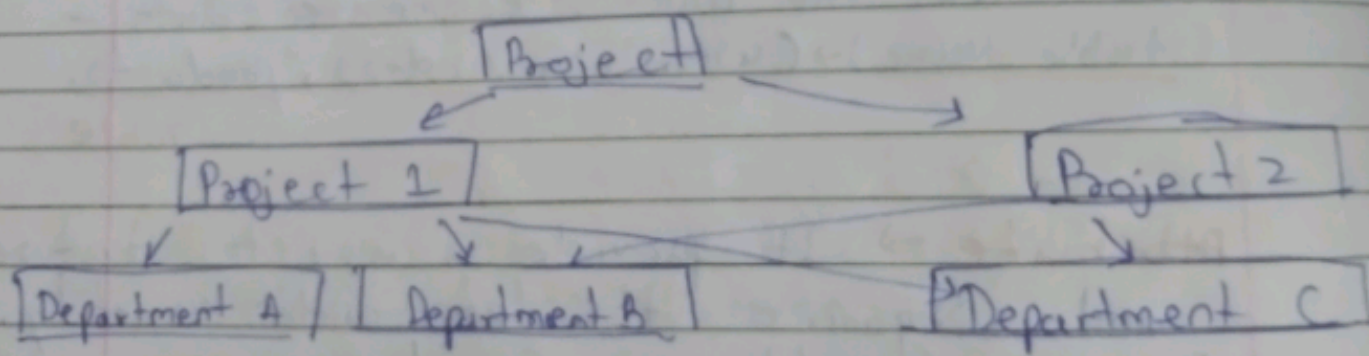
— many to many, many to one,
one to many, one to one

Relationship $\rightarrow$ Relationship define how entities are connected with each other.

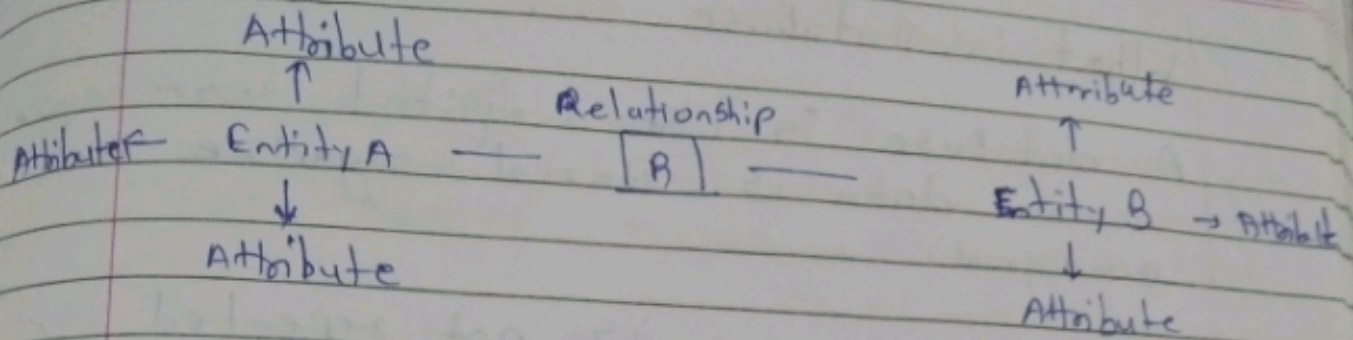
Reference Table - It is use to resolve the relationship
- Many to many, many to one,
one to many, one to one

Schema - A schema refers to the complete description or blueprints.

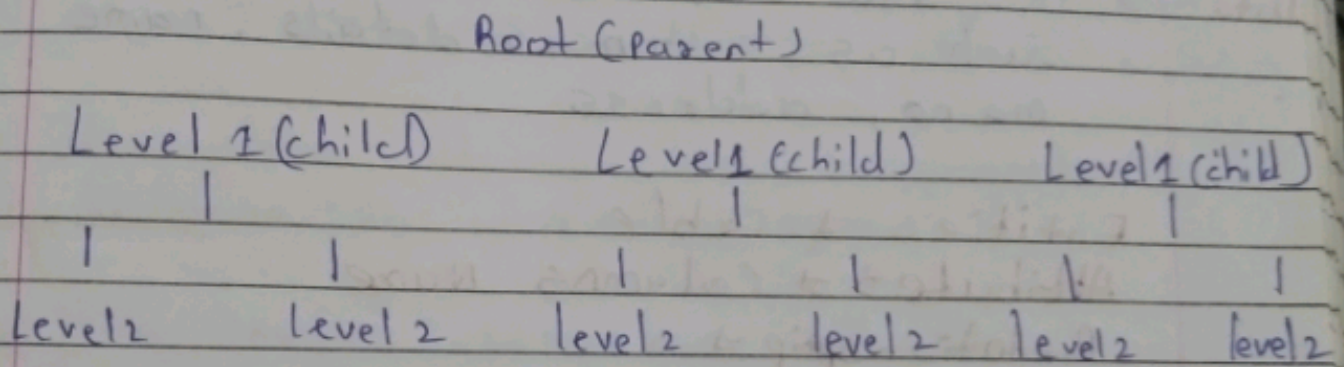
① Data model - Network Model



② E R MODEL $\rightarrow$ ENTITY RELATIONSHIP MODEL



③ Hierarchical Model [doubt]



④ Relational Model

Relational Model

Primary Key	Order Id	Customer Id	Order Date	Total Amount
	1001	1	2023-07-1	250
	1002	2	2023-07-2	150
	1003	3	2023-07-3	300
	1004	4	2023-07-4	200

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 What is Database ??

A database is like a digital warehouse where data is stored, organized and managed.

→ Database name is not repeated or same. Once created it's final.

Data → The raw information stored in a database such as customers details, name, no no, address

Entities → Table

Attributes → Columns Name

Relationship →

Queries — Queries are like a search commands that allow users to extract a specific data from the database, search, filter, update, Delete, Insert.

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 Different Type of Database.

- 1) Relational DataBase - It separate into rows and columns form in the table. SQL, ORACLE, MYSQL, MICROSOFT SQL.
- 2) NO SQL DataBase - No relational database - (Wikipedia) - MONDGO DB, CASSENDRA, COUCHBASE
- 3) OBJECT ORIENTED DATABASE - objectDb, db4o
- 4) Hierarchical Database - MIS - Management Information System - hospital.
- 5) Time series Database - InfluxDB,

RDBMS - RELATIONAL DATABASE MANAGEMENT SYSTEM

DBMS - DATABASE MANAGEMENT SYSTEM

MySQL - Relationship

- One to one
you are having adhar card, from adhar office
- one to many
- From Amazon one customer can purchase the multiple / many products
- many to one
- Hospital Database - One Surgeon can performs many operations
- many to many
- many Students can select the many subjects.

SQL - Command

Keys -

Diagram illustrating key types and their relationship to the Employee Table:

- Super Key**: Points to Candidate Key, Unique Key, and Alternate Key.
- Candidate Key**: Points to Primary Key, Unique Key, and Alternate Key.
- Primary Key**: Points to the Employee Table.
- Unique Key**: Points to the Employee Table.
- Alternate Key**: Points to the Employee Table.

Emp Id	Employee Name	Address	Licence	Passport
100	Jack	New Delhi	DL123456	PN456
101	John	Mumbai	MD99882	GAL278
102	Smith	Chennai	CH223899	ZZ4A7

Diagram illustrating key types and their relationship to the Salary Table:

- Composite Key**: Points to the Salary Table.
- Foreign Key**: Points to the Employee Table.

Emp Id	Salary Month Year	Amount
100	2015-Jan	10000
101	2015-Jan	20000